Grade 7 Mathematics

Strand 1: Number

Sub-strand 4: Ratios and Proportion

Lesson 1 — Ratio Language, Unit Rate, and Proportional Reasoning

Codes: B7.1.4.1.1 – B7.1.4.1.5

Teaching order inside this block: ratio language → unit rate → tables of equivalent ratios → proportional reasoning (missing values) → percent as a rate per 100.

1. What this lesson is about

Ratio compares two quantities: "3 pens to 5 pencils" is the ratio 3 : 5.

Proportion says two ratios are equal: $(3)/(5) = (6)/(10)$.

Rate compares different units: 20 km per hour, GH■ 3 per pen.

Words to unpack first:

Word	Meaning
Ratio	comparison of two quantities, same units
Unit rate	rate for **one** item (e.g. km per 1 hour)
Equivalent ratio	same relationship, scaled up/down
Proportional	one quantity grows at the same rate as the other
Rate per 100	another way to say percentage

2. Ratio language (B7.1.4.1.1)

- Write ratios with `:` (e.g. **3 : 5**) or as fractions ($(3)/(5)$).
- **Simplify** by dividing both sides by the HCF: $18 : 24 = 3 : 4$.
- Order matters: 3 : 5 is **not** the same as 5 : 3.
- **Part to part** vs **part to whole**: if the ratio is 3 : 5, total parts = 8.
- Part to part: 3 : 5
- Part to whole: 3 : 8 and 5 : 8.

Example: In a class of 40, boys : girls = 3 : 5. Total parts = 8. Boys = $(3)/(8) \times 40 = 15$; girls = 25.

3. Unit rate (B7.1.4.1.2)

A **unit rate** is "per one".

Example: 250 km in 5 hours → $250 \div 5 = 50$ km per hour.

Example: 8 pens cost GH■ 24 → GH■ 3 per pen.

4. Tables of equivalent ratios (B7.1.4.1.3)

Scale a ratio up and down to build a table.

Example: 2 : 3 (apples : oranges)

Apples	2	4	6	8
Oranges	3	6	9	12

Each row is an **equivalent ratio** ((2)/(3) = (4)/(6) = (6)/(9) = (8)/(12)).

5. Proportional reasoning — missing values (B7.1.4.1.4)

If two ratios are equal, use **cross-multiplication** or **scale factor**.

Scale factor: 2 : 3 → 6 : 9 (× 3).

Cross-multiplication: (a)/(b) = (c)/(d) ⇒ a × d = b × c

Example: (x)/(6) = (3)/(4) 4x = 18 \Rightarrow x = 4.5.

6. Percent as a rate per 100 (B7.1.4.1.5)

30% of a quantity = 30 out of 100 = 0.3 × the quantity.

Example: 30% of 250 (30)/(100) × 250 = 75

Share in a ratio: share 720 in 2 : 3. Total parts = 5; one part = 144; shares = 288 and 432.

7. Twenty fully worked examples

1. Simplify 18 : 24 → 3 : 4.
2. Boys : girls = 3 : 5, class of 40 → boys 15, girls 25.
3. Ratio 2 : 3 as fractions of total → (2)/(5) and (3)/(5).
4. 250 km in 5 h → unit rate 50 km/h.
5. 8 pens for GH■ 24 → GH■ 3 each.
6. Table 2 : 3 → 4 : 6, 6 : 9, 8 : 12.
7. (x)/(6) = (3)/(4) → x = 4.5.
8. 30% of 250 → 75.
9. Share 720 in 2 : 3 → 288 and 432.
10. Share 900 in 2 : 3 : 4 → 200, 300, 400.
11. y \propto x, y = 15 when x = 5 → k = 3; y = 24 when x = 8.
12. 5 pens cost GH■ 17.50 → 8 pens cost GH■ 28.
13. Recipe for 4 uses 300 g → for 10 uses 750 g.
14. 3 : 5 ratio, total 96 → 36 and 60.
15. Map 1 : 50 000, 4 cm → 2 km.
16. 20% of 400 → 80.
17. Which is larger: (2)/(3) or (3)/(5)? → (2)/(3).
18. Unit rate: 120 km in 3 h → 40 km/h.
19. (2)/(3) = (x)/(9) → x = 6.
20. Share 720 in 1 : 2 → 240 and 480.

8. Eight exercise sets × 5 questions

Exercise 1 — Ratio language

1. Simplify 18 : 24.
2. Ratio 2 : 3 as two fractions of the total.

3. Boys : girls = 3 : 5, class of 40. Find boys.
4. Write the ratio of 3 pens to 5 pencils.
5. **[Review]** `(3)/(8) + (1)/(4)`.

Exercise 2 — Unit rate

1. 250 km in 5 hours. Find km/h.
2. 8 pens cost GH■ 24. Cost of one pen.
3. 120 km in 3 hours. Find km/h.
4. 45 words typed in 5 minutes. Words per minute.
5. **[Review]** `2^3 × 2^4`.

Exercise 3 — Tables of equivalent ratios

1. Complete: 2 : 3 → 4 : 6 → 6 : 9 → 8 : ?
2. Complete: 5 : 2 → 10 : 4 → 15 : ?
3. Is 6 : 9 equivalent to 2 : 3?
4. Write a ratio equivalent to 4 : 5.
5. **[Review]** Order `(3)/(8), (1)/(2), (5)/(16)`.

Exercise 4 — Proportional reasoning (missing values)

1. `(x)/(6) = (3)/(4)`. Find x.
2. `(2)/(3) = (x)/(9)`. Find x.
3. `y \propto x`; `y = 15` when `x = 5`. Find y when `x = 8`.
4. 5 pens cost GH■ 17.50. Cost of 8 pens.
5. **[Review]** `124.3 + 0.275 + 74.06` to 1 d.p.

Exercise 5 — Sharing in a ratio

1. Share GH■ 720 in 2 : 3.
2. Share GH■ 900 in 2 : 3 : 4.
3. Two numbers in ratio 3 : 5, sum 96. Find them.
4. Share 720 in 1 : 2.
5. **[Review]** Value of 7 in 832 713.

Exercise 6 — Percent as rate per 100

1. 30% of 250.
2. 20% of 400.
3. 35% of GH■ 240.
4. 15% discount on GH■ 80. Find the sale price.
5. **[Review]** `(2)/(5) ÷ (14)/(15)`.

Exercise 7 — Real-world rates & scales

1. Recipe for 4 uses 300 g. For 10 people?
2. Map 1 : 50 000. What does 4 cm represent in km?
3. Which is larger: `(2)/(3)` or `(3)/(5)`?
4. 120 km at 40 km/h. Time taken.
5. **[Review]** Round 12.473 to 2 d.p.

Exercise 8 — Mixed / challenge

1. `(2)/(3) = (x)/(9)`. Find x.
2. Share 720 in 1 : 2.

3. $y = kx$; $y = 21$ when $x = 7$. Find y when $x = 10$.
4. **[Review]** $(3)/(8)$ of 480.
5. **[Review]** $2^3 \times 2^4$.

9. Full worked solutions

Solutions — Exercise 1

1. **3 : 4**. 2. $(2)/(5)$ and $(3)/(5)$. 3. **15 boys**. 4. **3 : 5**. 5. $(5)/(8)$.

Solutions — Exercise 2

1. **50 km/h**. 2. **GH■ 3**. 3. **40 km/h**. 4. **9 words/minute**. 5. **128**.

Solutions — Exercise 3

1. **12**. 2. **6**. 3. **Yes** ($(6)/(9) = (2)/(3)$). 4. e.g. **8 : 10**. 5. $(5)/(16) < (3)/(8) < (1)/(2)$.

Solutions — Exercise 4

1. $x = 4.5$. 2. $x = 6$. 3. $k = 3$; $y = 24$. 4. **GH■ 28**. 5. **198.6**.

Solutions — Exercise 5

1. **288** and **432**. 2. **200, 300, 400**. 3. **36** and **60**. 4. **240** and **480**. 5. **700**.

Solutions — Exercise 6

1. **75**. 2. **80**. 3. **GH■ 84**. 4. **GH■ 68**. 5. $(3)/(7)$.

Solutions — Exercise 7

1. **750 g**. 2. **2 km**. 3. $(2)/(3)$. 4. **3 hours**. 5. **12.47**.

Solutions — Exercise 8

1. $x = 6$. 2. **240** and **480**. 3. $k = 3$; $y = 30$. 4. **180**. 5. **128**.

10. Cheat sheet

Skill	Rule
Simplify ratio	divide both by HCF
Unit rate	total ÷ number of items
Equivalent ratio	scale both sides the same
Missing value	cross-multiply or scale
Percent of quantity	$(p)/(100) \times$ quantity
Share in ratio	total ÷ parts, then multiply

NaCCA CCP B7.1.4 — plain English, British notation.